

Real Analysis

Quiz 1

Time Allowed: 45 Minutes**Maximum Marks:** 20

Instructions: Answer **all** questions. Marks are indicated against each question.

Q1. Use the **Principle of Mathematical Induction** to prove the following: [5 Marks]

(a) Prove that $5^{2n} - 4n - 1$ is divisible by 16, for all $n \in \mathbb{N}$.

(b) Prove that

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} > \sqrt{n}, \quad \text{for all } n > 1.$$

Q2. For $A \subseteq \mathbb{N}$, define $f_A : \mathbb{N} \rightarrow \mathbb{R}$ by

[4 Marks]

$$f_A(n) = \begin{cases} 1 & \text{if } n \in A, \\ 0 & \text{if } n \notin A. \end{cases}$$

Using **Cantor's Theorem**, show that the set $\{f_A : A \subseteq \mathbb{N}\}$ is **uncountable**.

Q3. Let $f : A \rightarrow B$ be a function and let G, H be subsets of B . Show that: [4 Marks]

$$\begin{aligned} f^{-1}(G \cup H) &= f^{-1}(G) \cup f^{-1}(H), \\ f^{-1}(G \cap H) &= f^{-1}(G) \cap f^{-1}(H). \end{aligned}$$

Q4. Prove that for all $x, y, z \in \mathbb{R}$,

[3 Marks]

$$|x| + |y| + |z| \leq |x + y - z| + |y + z - x| + |z + x - y|.$$

Q5. Let y be a positive real number. Show that there exists a natural number m such that [4 Marks]

$$0 < \frac{1}{2^m} < y.$$

— End of Question Paper —